

HAT-P-36b

R-1945

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250616 · created 2026-08-06T15:00:10+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460842.7022 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.033 +/- 0.028
Transit depth (Rp/Rs)^2	0.11 +/- 0.18 [%]
Orbital Inclination (inc)	80.41 +/- 2.83
Airmass coefficient 1 (a1)	1.213 +/- 0.02
Airmass coefficient 2 (a2)	-0.142 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.66 %
Best Comparison Star	None
Optimal Aperture	3.89
Optimal Annulus	12.64
Transit Duration (day)	0.054 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.033 ± 0.028 vs 0.126 (deviation 3.13σ)
Duration fit vs published	0.054 d vs 0.0925 d (deviation 1.73σ)
Mid-time O–C	-15.4 min
Depth SNR	1.1
Residual scatter	1.66 %

Light curve

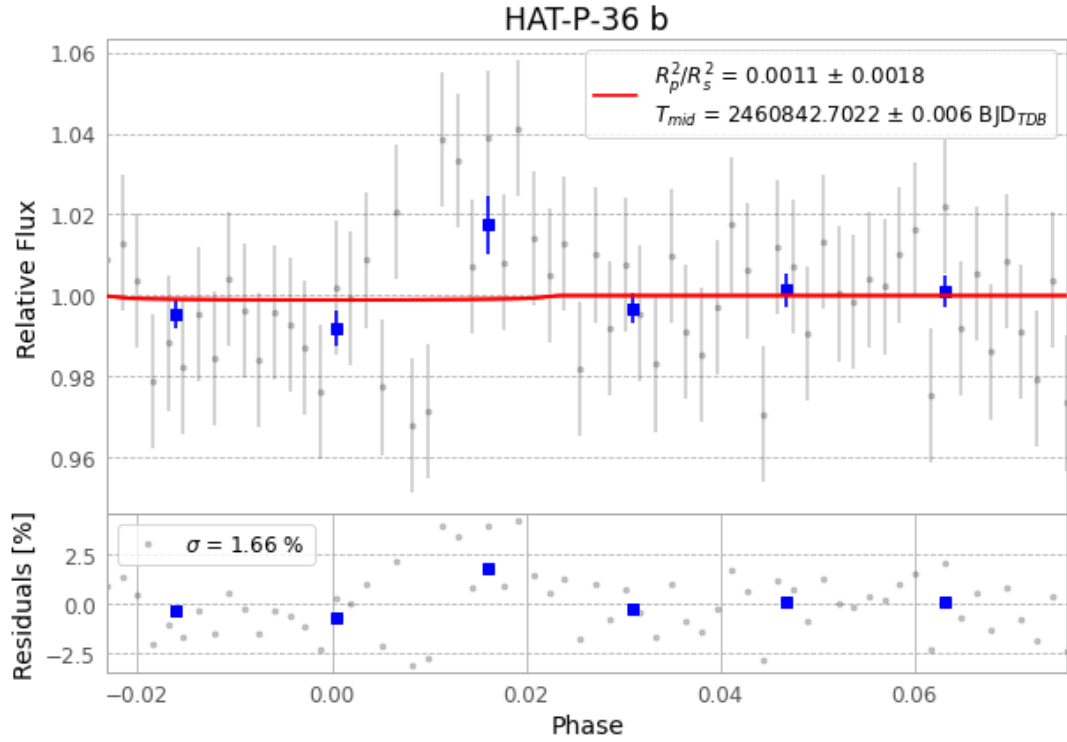


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-06-15.png (SHA-256 9cd718c9568ce1f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 253], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 89d8791e1263568b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), HATP-36250616040615.FITS → HATP-36250616071516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250616.log (dde51022d6f9...), FinalLightCurve_HAT-P-36 b_2025-06-15.png (9cd718c9568c...), FinalLightCurve_HAT-P-36 b_2025-06-15.csv (c67a25063858...)

Compute resources

Wall time 128.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:00Z.